

Data Generation

STEP 1. Cluttered Scene Generation

Object assets



1,148 objects

Scene assets



100 scenes
(13 categories)

6K cluttered scenes



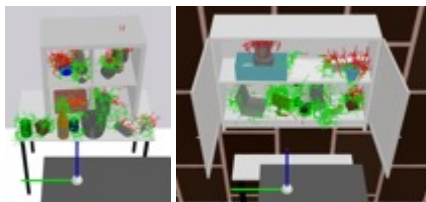
Low

Mid

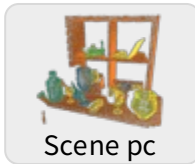
High

STEP 2. Obstruction Labeling

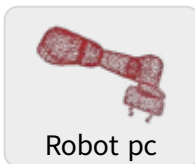
Sample grasp poses + Solve IK



Obstruction Dataset



Scene pc



Robot pc



object-wise obstruction labels

Grasp Obstruction Reasoning

Scene pc



Robot pc g_0



VORM

Per-grasp Obstruction Prediction



grasp pose g_0

Obstruction set
: {fixed object, box}

Feasible ✗



grasp pose g_1

Obstruction set
: {bowl}

Feasible ✗



grasp pose g_2

Obstruction set : None

Feasible ✓

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